Drifting as a Flow Map

Abstract

We study learned two-time maps for generation while separating three methods that share notation but not objectives. Our historical WGF core combines an origin endpoint-drifting update, two finite-difference kernel-WGF residuals, and detached canonical composition. A source-locked CelebA-64 confirmation of that core gives FID 45.762 ± 1.513 at four network evaluations, versus $61.556 \pm .590$ at one, over three training seeds. Our newer pure Lagrangian-self-distillation (LSD) baseline instead adapts the released flow-map objective with an exact time-JVP and no WGF, GMD, endpoint teacher, or separate semigroup penalty. At its pre-registered 100k endpoint, raw weights give FID 34.377 at one evaluation and 30.799 at sixteen; the corresponding EMA sensitivity values are 32.358 and 27.613. A pre-registered D2 pilot resumes the exact raw 30k state and adds a weak, linearly ramped terminal GMD anchor through 40k. Against its paired pure-LSD continuation, D2 raw improves FID from 37.625 to 33.403 at one evaluation and from 31.503 to 30.404 at sixteen, satisfying every registered pilot criterion. A later half-dose D4 ablation also improves raw FID throughout, but its EMA at sixteen exceeds pure LSD by 0.00654 FID and 1.41×10^{-5} KID; its strict conjunctive verdict is therefore `NO_GO`. These LSD, D2, and D4 values, like the latent GMD-v1 (37.413) and latent GMD-v2 (39.379), are single-seed development measurements. A pre-registered one-factor RGB screen keeps the same MAE-only GMD recipe and adds 13 frozen ConvNeXt-V2 families; at 10k/EMA/NFE one its FID falls from 53.262 to 18.887 and its KID from 44.640 to 8.841 in 10^{-3} units, while measured training time per update increases by a factor 4.07. These RGB values also use the reused 4,999-image holdout and are not population comparisons with the historical three-seed table. The evidence supports specific learned-map implementations, a promising paired D2 pilot, a narrowly negative D4 dose ablation, and a strong local feature intervention—not a certified WGF solver or a general method-wide claim.

1 The object: a learned flow map

Let p_0 be a simple prior on $\mathbb{R}^d$ and p a target law. Begin with a two-time transition family

$$\Phi_\theta : \mathbb{R}^d \times \{(a, b) : 0 \le a \le b \le 1\} \to \mathbb{R}^d. \tag{1}$$

An exact non-autonomous flow (or evolution family) would additionally satisfy

$$\Phi(x, a, a) = x, \qquad \Phi(\Phi(x, a, b), b, c) = \Phi(x, a, c) \quad (0 \le a \le b \le c \le 1). \tag{2}$$

We use "learned flow map" as shorthand for a transition family with the first identity enforced architecturally and the second, cocycle identity learned approximately on sampled model-reached states. We do not assume a global cocycle before training. It induces the canonical path

$$X_t^\theta(z) = \Phi_\theta(z, 0, t), \qquad q_t^\theta = (X_t^\theta)_{\#} p_0. \tag{3}$$

At inference, a partition $0 = t_0 < \cdots < t_K = 1$ gives

$$x_0 = z \sim p_0, \qquad x_{k+1} = \Phi_\theta(x_k, t_k, t_{k+1}). \tag{4}$$

Thus optimizer time is not the only notion of motion: the trained model itself represents transitions along a normalized time coordinate.

This object differs from continuous normalizing flow matching, which regresses a vector field along a prescribed probability path [4], and from consistency models, which directly train one- and few-step mappings along diffusion trajectories [5]. Here the path is induced by the same learned two-time transition family, and NFE changes how that family is composed rather than the discretization of a fixed learned ODE.

The closest precedents delimit the claims made here. Boffi, Albergo, and Vanden-Eijnden introduced direct flow-map learning by self-distillation, including the Lagrangian objective adapted below [6]. Deng et al. introduced Generative Modeling via Drifting (GMD) and its feature-space distributional loss [7]. Li and Zhu already gave a semigroup-consistent long–short flow-map interpretation of drifting models [8]. Accordingly, neither the generic connection between drifting and flow maps nor LSD itself is claimed as new here. The historical WGF construction, the released-objective adaptations, and the local CelebA measurements are kept distinct throughout.

The historical WGF-core latent implementation used for CelebA enforces the identity boundary algebraically:

$$\alpha(a,b) = \begin{cases} (b-a)/(1-a), & a < 1, \\ 0, & a = b = 1, \end{cases} \tag{5}$$

$$\Phi_\theta(x,a,b) = (1-\alpha(a,b))x + \alpha(a,b)g_\theta(x,a,b). \tag{6}$$

Consequently $\Phi_\theta(x,a,a) = x$ exactly, and every nontrivial transition with $a < 1$ that ends at 1 is a learned endpoint. The endpoint network still depends on its input; "endpoint" does not mean a constant output. Along any partition, the explicit coefficient of the original source telescopes:

$$\prod_{k=0}^{K-1}(1-\alpha(t_k,t_{k+1})) = \prod_{k=0}^{K-1}\frac{1-t_{k+1}}{1-t_k} = 1-t_K. \tag{7}$$

In particular, the explicit source skip vanishes at $t_K = 1$ for every partition. This parameterization, rather than a claimed ODE solver, is the simple architectural meaning of "drifting as a flow map."

These are architectural identities, not learned consequences of a loss. The legacy MNIST and pixel implementations use the same expression with a 10^{-3} floor on $1-a$. Their diagonal remains an exact identity, but their explicit endpoint skip is not guaranteed to vanish for starts within 10^{-3} of time one. We do not use that numerical guard in the historical latent WGF implementation and do not promote the legacy approximation to a general theorem.

2 Historical WGF teacher field

This section records the WGF-core method used in the historical MNIST and CelebA experiments. It is not the objective of the modern LSD or GMD runs reported later.

For motivation, let $\mathcal{F}_p(q)$ be an energy whose Wasserstein gradient flow satisfies [2]

$$\partial_s q_s + \nabla\cdot(q_s V_{p,q_s}) = 0, \qquad V_{p,q}(x) = -\nabla_x \frac{\delta \mathcal{F}_p}{\delta q}(x). \tag{8}$$

We use normalized model time $t \in [0,1]$ and physical time $s = \sigma(t)$. Experiments use the constant schedule $\sigma(t) = St$, so a normalized step ε represents a physical step $\Delta\sigma = S\varepsilon$.

2.1 KL–KDE velocity

For $\mathcal{F}_p(q) = \mathrm{KL}(q\|p)$, the formal velocity is

$$V_{p,q}(x) = \nabla \log p(x) - \nabla \log q(x). \tag{9}$$

Given a support $U = (u_j)_{j=1}^n$, the raw-coordinate Gaussian KDE estimator [1] gives the score used by the implementation:

$$w_j(x; U, h) = \frac{\exp[-\|x - u_j\|^2 / (2h^2)]}{\sum_\ell \exp[-\|x - u_\ell\|^2 / (2h^2)]}, \tag{10}$$

$$\widehat{s}_h(x; U) = \frac{1}{h^2} \sum_j w_j(x; U, h)(u_j - x), \tag{11}$$

$$\widehat{V}(x; Y, R, h) = \widehat{s}_h(x; Y) - \widehat{s}_h(x; R), \tag{12}$$

where Y is a data support and R an independent model support. Logits are normalized with a softmax, avoiding a division by an underflowed kernel sum. The bandwidth is nevertheless a statistical and physical hyperparameter: numerical stability alone does not make a poor bandwidth informative. Like Stein variational gradient descent [3], this construction transports empirical particles with a kernelized field, but Equation (12) is a plug-in difference of KDE scores rather than the RKHS-optimal SVGD direction.

Equation (9) is exact only for population scores. Equation (12) is a finite-particle, smoothed approximation. All claims below concern the implemented estimator, not exact integration of the KL gradient flow.

2.2 Raw and dimension-normalized units

The code exposes two RBF geometries. In *raw* geometry the exponent uses $\|x - u\|^2$. In *dimension-normalized* geometry it uses $\|x - u\|^2 / d$, but deliberately keeps h^2, rather than dh^2, in the velocity denominator. Hence the exact implementation-level conversion is

$$\widehat{V}_{\text{norm}}(h) = d\, \widehat{V}_{\text{raw}}(\sqrt{d}\, h). \tag{13}$$

To compare the same update in the two conventions, one must therefore set

$$h_{\text{norm}} = h_{\text{raw}}/\sqrt{d}, \qquad S_{\text{norm}} = S_{\text{raw}}/d, \qquad \eta_{\text{norm}} = \eta_{\text{raw}}/d, \qquad \lambda_{\text{W,norm}} = \lambda_{\text{W,raw}}/d^2, \tag{14}$$

where η is the endpoint drifting step and λ_{W} weights the two squared path-WGF residuals in Equation (24). Indeed, the velocity and the finite-difference quotient are both multiplied by d after the simultaneous S conversion, so their mean-squared residual is multiplied by d^2. Equivalently, one may divide the product $\lambda_{\text{path}}\lambda_{\text{W}}$ by d^2. At $d = 256$, raw $(h, S, \eta, \lambda_{\text{W}}) = (4, 100, 1, 1)$ corresponds to normalized $(0.25, 0.390625, 1/256, 1/65536)$. Changing only the bandwidth is not a controlled geometry ablation. This RBF-unit conversion is unrelated to the channelwise affine normalization introduced below.

2.3 Global particle support under DDP

With W distributed ranks, local batch size B, and G contiguous time groups, a rank-local query group contains B/G points. A rank-local KDE uses B data supports and B/G model-reference supports at that group's time. Our global option all-gathers detached supports: the data KDE then uses WB points and each model-reference chunk uses WB/G points. When synchronized times are also enabled, as in the DDP protocol below, the G times are broadcast from rank zero and the gathered model tensor is reordered group-first. Its gth chunk then contains time a_g from every rank. This qualification matters: without synchronized times, a gathered chunk would mix different q_a^θ; its KDE would estimate a mixture of times rather than the score at one queried time.

The gathered supports and teacher velocities are detached. DDP therefore increases the particle support without introducing cross-rank autograd through the KDE teacher; larger support need not by itself improve the learned model.

3 Historical WGF-core training without rollout or time-JVP

All teacher quantities in this section are stop-gradient quantities. This is a semi-gradient algorithm: at every optimizer step, the current model defines fresh samples and targets, but differentiation treats those targets as constants. Student branches still receive ordinary backpropagation. For exact agreement with the implementation, each notation $\mathbb{E}\,\|r\|^2$ below denotes mean squared error over both the batch and the d coordinates, namely $\mathbb{E}[\|r\|_2^2/d]$.

3.1 Two-view finite-difference WGF block

Sample $z, \tilde{z} \sim p_0$, a time $a < 1 - 2\varepsilon$, and set

$$x_a = \mathrm{sg}(\Phi_\theta(z, 0, a)), \qquad r_a = \mathrm{sg}(\Phi_\theta(\tilde{z}, 0, a)), \qquad v_a = \mathrm{sg}(\widehat{V}(x_a; Y, r_a, h)). \tag{15}$$

The implementation uses both an origin-path finite difference and a local transition finite difference:

$$\mathcal{L}^0_{\mathrm{WGF}} = \mathbb{E}\left\| \frac{\Phi_\theta(z, 0, a+\varepsilon) - x_a}{\Delta\sigma} - v_a \right\|^2, \tag{16}$$

$$\mathcal{L}^{\mathrm{loc}}_{\mathrm{WGF}} = \mathbb{E}\left\| \frac{\Phi_\theta(x_a, a, a+\varepsilon) - x_a}{\Delta\sigma} - v_a \right\|^2, \tag{17}$$

$$\mathcal{L}_{\mathrm{WGF}} = \mathcal{L}^0_{\mathrm{WGF}} + \mathcal{L}^{\mathrm{loc}}_{\mathrm{WGF}}. \tag{18}$$

This block supplies a local directional preference. The selected core uses both views; the experiments do not isolate either residual individually. The empirical evidence also does not support presenting their finite-batch target as an accurately solved PDE constraint.

3.2 Canonical flow-map composition

For a sampled $a < b$, the core consistency target is

$$\mathcal{L}_{\mathrm{S}} = \mathbb{E}\,\|\Phi_\theta(\mathrm{sg}[\Phi_\theta(z, 0, a)], a, b) - \mathrm{sg}[\Phi_\theta(z, 0, b)]\|^2. \tag{19}$$

The direct origin path is a detached teacher and the second transition is the student. Equation (19) asks for the flow-map identity only on states reached from p_0 and sampled time pairs; it is not a global cocycle/composition guarantee for every $x \in \mathbb{R}^d$. Its role is simple: the map used from a to b should agree with the origin-defined path that supplies q_b^θ. The implementation samples G grouped start times uniformly on $[0, 1-2\varepsilon)$ and sets $b = a + \varepsilon + u\min(1 - a - 2\varepsilon, m)$, with $u \sim \mathcal{U}[0,1]$. The additional span m increases linearly from .25 to .5 over training.

3.3 Optional Eulerian regularization

For $a + \varepsilon \le b < 1$, define

$$x_a^+ = \Phi_\theta(x_a, a, a+\varepsilon), \qquad x_b = \Phi_\theta(x_a, a, b). \tag{20}$$

The Eulerian composition loss is

$$\mathcal{L}_{\mathrm{E}} = \mathbb{E}\left\| \Phi_\theta(x_a^+, a+\varepsilon, b) - \mathrm{sg}(x_b) \right\|^2. \tag{21}$$

Within $\mathcal{L}_{\mathrm{E}}$, gradients pass through the two composed student transitions, while the direct map is detached. The composed branch is a two-transition prefix of Equation (4); the loss is weaker than asserting an exact composition identity everywhere. It is disabled in the minimal core and added only in the Eulerian-regularized variant.

3.4 Origin endpoint drifting

The origin endpoint and its independent reference are

$$y_0 = \Phi_\theta(z, 0, 1), \qquad \tilde{y}_0 = \mathrm{sg}(\Phi_\theta(\tilde{z}, 0, 1)). \tag{22}$$

One detached WGF step defines the target

$$\mathcal{L}_{\mathrm{B}}^0 = \mathbb{E}\left\| y_0 - \mathrm{sg}[y_0 + \eta \widehat{V}(y_0; Y, \tilde{y}_0, h)] \right\|^2. \tag{23}$$

This is the direct flow-map analogue of ordinary drifting.

3.5 Historical selected objective

The minimal active objective selected for the historical CelebA WGF study is

$$\boxed{\mathcal{L}_{\mathrm{core}} = \lambda_{\mathrm{path}} \lambda_{\mathrm{W}} \mathcal{L}_{\mathrm{WGF}} + \lambda_{\mathrm{S}} \mathcal{L}_{\mathrm{S}} + \lambda_0 \mathcal{L}_{\mathrm{B}}^0.} \tag{24}$$

The selected coefficients are

$$(\lambda_{\mathrm{path}}, \lambda_{\mathrm{W}}, \lambda_{\mathrm{S}}, \lambda_0) = (0.1, 0.25, 0.1, 5), \qquad (\lambda_{\mathrm{E}}, \lambda_c, \lambda_{\mathrm{L}}) = (0, 0, 0). \tag{25}$$

Thus the effective weight of the two-view WGF block is 0.025. The Eulerian-regularized variant adds

$$\mathcal{L}_{\mathrm{reg}} = \mathcal{L}_{\mathrm{core}} + \lambda_{\mathrm{path}} \lambda_{\mathrm{E}} \mathcal{L}_{\mathrm{E}}, \qquad \lambda_{\mathrm{E}} = 100, \tag{26}$$

whose effective Eulerian weight is 10.

The three historical core ideas can therefore be read directly from Equation (24): endpoint drifting learns where to finish, the WGF block supplies a direction, and canonical composition makes transitions reusable. The arbitrary-start endpoint loss used in earlier experiments is, for a batch-shared $c \in [0, 1]$,

$$x_c = \mathrm{sg}(\Phi_\theta(z, 0, c)), \qquad y_c = \Phi_\theta(x_c, c, 1), \tag{27}$$

$$\tilde{x}_c = \mathrm{sg}(\Phi_\theta(\tilde{z}, 0, c)), \qquad \tilde{y}_c = \mathrm{sg}(\Phi_\theta(\tilde{x}_c, c, 1)), \tag{28}$$

$$\mathcal{L}_{\mathrm{B}}^c = \mathbb{E}\left\| y_c - \mathrm{sg}[y_c + \eta \widehat{V}(y_c; Y, \tilde{y}_c, h)] \right\|^2. \tag{29}$$

It uses a separate reference KDE rather than pairing y_0 with y_c. Its coefficient is zero in both selected variants. The historical Lagrangian finite-difference self-distillation coefficient is also zero. For legacy diagnostics the implementation still evaluates both zero-weight residuals whenever the origin boundary is active; they consume computation (and the arbitrary branch consumes random draws) but contribute exactly zero gradient.

No full inference partition from Equation (4) participates in the training loss or back-propagation; either composition term uses only two transitions. Periodic monitoring can run inference-only partition rollouts. There is no Jacobian–vector product and no mixed time–parameter derivative. This historical method uses ordinary forward passes through Φ_θ and detached particle velocities.

4 Modern latent LSD and the completed D2/D4 pilots

4.1 Pure latent LSD baseline

Our current pure LSD baseline is a PyTorch adaptation of the released Lagrangian flow-map objective of Boffi et al. [6], at release commit `2f115a07fa9073553193e4b265dfc303827af2b0`. It retains the released objective but substitutes the local frozen-VAE $4 \times 8 \times 8$ latent data, a

compact ordinary-convolution U-Net, and local distributed execution; it is not a replication of the paper's pixel-space EDM2 experiment.

For independent $x_0 \sim \mathcal{N}(0, I)$ and $x_1 \sim p_{\text{lat}}$, define

$$X_\theta(x, s, t) = x + (t - s)\phi_\theta(x, s, t), \qquad b_\theta(x, t) = \phi_\theta(x, t, t), \tag{30}$$

and use the linear stochastic interpolant $I_t = (1 - t)x_0 + tx_1$. The diagonal grounding residual and off-diagonal Lagrangian residual are

$$r_{\text{D}}(t) = b_\theta(I_t, t) - (x_1 - x_0), \tag{31}$$

$$r_{\text{L}}(s, t) = \partial_t X_\theta(I_s, s, t) - \text{sg}[b_\theta(\text{sg}[X_\theta(I_s, s, t)], t)], \qquad 0 \le s \le t \le 1. \tag{32}$$

The endpoint-time derivative is computed by an exact automatic-differentiation JVP with a graph retained for the student update. The stopped target uses the current raw parameters, not the EMA. A learned two-time log-variance $w_\theta(s, t)$ weights each per-sample squared residual as $\exp[-w_\theta(s, t)] \, \|r\|_2^2 + w_\theta(s, t)$. Each local batch is split exactly 75/25 between diagonal and off-diagonal terms, and those proportions also weight their means. The pure baseline contains no WGF/KDE block, GMD loss, endpoint-teacher update, separate composition loss, or inference rollout.

4.2 Standalone GMD adaptations

The standalone GMD controls use a PyTorch port of the released JAX loss of Deng et al. [7], reference commit `accd0cf09c33b70892d33941d2a287ca86cb92e1`. Golden JAX comparisons cover the loss and parameter gradients with and without fixed negatives, and a two-rank test covers distributed support, the first AdamW update, optimizer state, clipping, and EMA. These checks establish numerical agreement of the loss implementation under the tested inputs; they do not make the CelebA training recipe an official or bitwise-identical replication.

All local GMD runs are unconditional CelebA-64 adaptations evaluated as one-step generators. The latent v1 and v2 controls generate frozen-VAE $4 \times 8 \times 8$ latents with a compact 21M-parameter network; v2 uses the longer-trained local MAE-v2 and its adapted `official_70` feature grid. Both use global batch 512, AdamW, and temperatures $(0.2, 0.05, 0.02)$. The direct-RGB control instead maps Gaussian RGB noise directly to 64×64 pixels with a 131,990,832-parameter DiT-B. It has no VAE, uses 82 nondegenerate feature families from a frozen pixel MAE, global generated batch 64, and positive/negative banks of size 128/1000. A matched branch adds only 13 frozen ConvNeXt-V2 Base families. The latent, RGB, conditioning, architecture, support, and budget choices are explicit CelebA adaptations, so the latent and RGB controls are descriptive rather than matched ablations of one another; only the two RGB branches form a one-factor causal contrast.

The published GMD pixel ablation provides important context for the local direct-RGB result. On ImageNet-256, the authors' Pixel-B/16 ablation at 100 generator epochs improves from FID 32.11 with a weaker width-256 MAE trained for 192 epochs to 9.35 with a width-640 MAE trained for 1,280 epochs and classification fine-tuning, then to 3.70 when ConvNeXt-V2 features are also used [7]. With the stronger feature stack, longer generator training reaches 2.19 at 320 epochs and 1.76 at 640 epochs.

The local MAE-only direct-RGB screen instead uses a width-64 MAE trained for 10k updates, no classification fine-tuning, no ConvNeXt branch, and only 10k generator updates. Its FID 53.262 therefore showed that this deliberately under-budget adaptation was weak; it was not evidence that GMD intrinsically fails on CelebA. The subsequent pre-registered branch keeps the data order, random streams, DiT, support, banks, optimizer, and 10k budget fixed and adds only 13 frozen ConvNeXt-V2 Base families to the same 82 MAE families. At the unique EMA/NFE-one endpoint it obtains FID 18.8866 and KID 8.8412 in 10^{-3} units, versus 53.2621/44.6405 for MAE alone: a 64.540% relative FID reduction. Both pre-specified metric gates pass. Internal

Table 1: Published GMD Pixel-B/16 ImageNet-256 context. These FIDs use a different dataset, resolution, conditioning regime, sample count, and evaluation protocol; they are not numerical CelebA baselines.

Training representation	generator epochs	FID↓
MAE width 256, 192 MAE epochs	100	32.11
MAE width 640, 1,280 MAE epochs + classification	100	9.35
Previous row + ConvNeXt-V2	100	3.70
Strong feature stack	320	2.19
Strong feature stack	640	1.76

trainer time is 24,990.854 seconds rather than 6,142.319 seconds for 10k updates, or 2.4991 versus .6142 seconds per update (4.07×). Thus feature geometry is a demonstrated bottleneck in this local screen. The corresponding Slurm wall times are 7:01:09 and 1:45:28, including launcher and checkpoint overheads. Replication across seeds and the official CelebA split remain open; the published ImageNet numbers still are not quantitative CelebA baselines.

A separate post-hoc diagnostic reconstructs every tested family/temperature energy from the official affinities to within 1.2×10^{-8}. At the direct-RGB training support (64 generated queries, 128 positives, and 32 zero-weight fixed negatives), the mean effective attraction/repulsion neighborhoods across seven global or pooled families are 69.03/35.00, 14.66/9.06, and 14.03/10.70 at temperatures .2, .05, and .02. The official affinity floor supplies about 30% of the measured mass at .02. Replacing the generated queries with disjoint real images yields nearly the same statistics and field-energy ratios of only 1.18, 1.15, and 1.05. This gives no gross generator-specific kernel-collapse signal; it does not show equality of distributions or establish a cause of FID. The measurement is limited to quantized PNGs, static banks, and seven of 82 RGB families. Two latent spatial-standard-deviation families on 1×1 maps are exactly null and are retained in the frozen primary recipe.

4.3 D2 and D4: completed paired 40k pilots

D2 is a pre-registered local intervention that resumes the exact raw pure-LSD 30k training state and changes only the objective to

$$\mathcal{L}_{\mathrm{D2}}(k) = \mathcal{L}_{\mathrm{LSD}} + \lambda(k)\mathcal{L}_{\mathrm{GMD}}^{\mathrm{terminal}}, \qquad \lambda(k) = 0.00024191577475160165\,\frac{k - 30000}{10000}, \tag{33}$$

for the registered updates $k = 30001, \ldots, 40000$. The target coefficient is 10% of the raw-checkpoint equal-RMS diagnostic 0.0024191577475160164; that local gradient calibration is a guardrail, not a theoretical optimum or a predicted sample-quality gain. The terminal term uses the PyTorch port of the released GMD loss [7], the frozen 30k MAE-v2 `official_70` EMA features, temperatures $(0.2, 0.05, 0.02)$, and local support 32. Its planned primary comparison is against a paired pure-LSD continuation from the same state; raw weights are primary, EMA is a sensitivity analysis, and one- and sixteen-evaluation quality must both be retained. Both branches reached exactly 40k, and the immutable 20-cell aggregate passed its provenance, finiteness, covariance, and zero-overlap checks. D2 passed every pre-specified numerical criterion; the full paired curves are reported in Table 8.

The recorded post-30k training segments take .1833 seconds per update for pure LSD and .3684 seconds per update for D2 on eight MI210s, a ratio of 2.01×. This is an observed end-to-end run ratio, not an isolated kernel benchmark: D2 additionally executes the frozen feature encoder, constructs GMD supports, and differentiates the terminal distributional term. Adding this weak terminal loss is not the long–short construction of Li and Zhu [8], and D2 is not claimed as a replication or test of Long–Short Guidance.

After D2 was closed, D4 was separately pre-registered as a one-factor half-dose ablation. It restarts the same raw 30k checkpoint and changes only the terminal coefficient from 10% to 5% of the fixed equal-RMS calibration,

$$\lambda_{\mathrm{D4}}(k) = 0.00012095788737580082\,\frac{k-30000}{10000}. \tag{34}$$

It retains the same data, optimizer and RNG states, 70 frozen MAE feature families, local support 32, and evaluation grid. D4 took .3300 seconds per observed update, or 1.80× pure LSD. Because multiplying the loss by one half removes no forward or backward operation, the difference from D2's .3684 is treated as run-to-run systems variation, not as a causal speedup.

5 What the historical WGF objective does and does not imply

Proposition 1 (Population idealization)**.** *Suppose the identity boundary gives $X_0^\theta(z) = z$, the population velocity in Equation* (8) *is smooth with unique characteristics, $t \mapsto X_t^\theta(z)$ is almost surely differentiable, and differentiation may pass under the expectation below. If the infinitesimal versions of Equation* (18) *vanish along the model path and exact composition holds on all visited states, then q_t^θ follows the time-rescaled WGF and Φ_θ agrees with its characteristic transition map on those states.*

Proof. The first condition gives $\partial_t X_t^\theta(z) = \sigma'(t) V_{p,q_t^\theta}(X_t^\theta(z))$. Differentiating $\mathbb{E}[\varphi(X_t^\theta(z))]$ for a smooth test function φ gives the weak continuity equation. Exact composition then maps $X_a^\theta(z)$ to $X_b^\theta(z)$ for every visited characteristic. □

This proposition interprets the historical WGF objective in a population, infinitesimal limit; it is not an empirical theorem about a trained network. In practice we use a finite bandwidth, finite support, finite differences, stop-gradients, and an imperfect composition loss. The endpoint terms are finite drifting updates, not proof that $q_1^\theta = p$ or that q_1^θ is a stationary point.

Nor should increasing NFE be described as reducing the discretization error of a fixed learned vector field. Different intervals are different neural conditionings. In the exact composition (cocycle) limit every partition produces the same endpoint; away from that limit, more compositions may improve, preserve, or damage quality. In our sampler, NFE $= K$ means exactly K calls to the transition network on the partition $t_k = (k/K)^\gamma$; NFE one is the direct endpoint $\Phi_\theta(z, 0, 1)$. An NFE trend therefore measures the empirical effect of reusing the learned conditional transitions, not convergence of a numerical ODE solver.

The reported transition diagnostic is also narrower than sample quality. On canonical states $X_{t_i}^\theta(z)$ it measures, for $t_i < t_j$,

$$\frac{\mathrm{RMSE}\Big(\Phi_\theta(X_{t_i}^\theta(z), t_i, t_j) - X_{t_j}^\theta(z)\Big)}{\mathrm{RMS}\Big(X_{t_j}^\theta(z) - X_{t_i}^\theta(z)\Big)}. \tag{35}$$

It tests one direct transition against the origin-defined canonical path; it does not compare distributions to data and does not certify an entire K-step rollout. We therefore report quality at each reported NFE and give the transition diagnostic separately; we never use a small composition error by itself as evidence of sample quality.

6 CelebA in a frozen latent space

Raw-coordinate pixel KDE was ineffective in the tested historical screen; this does not imply that every pixel-space distributional loss must fail. For the latent CelebA-64 studies, we instead freeze the `stabilityai/sd-vae-ft-mse` encoder and decoder, write μ_{VAE} for the encoder's

posterior mean and D_{VAE} for the decoder [14], following the general latent-space strategy of latent diffusion models [9]. Images are center-cropped, resized to 64×64, mapped from $[0,1]$ to the VAE input range $[-1,1]$, and encoded using the posterior mean, with no latent sampling. This sampling-free rule does not imply bitwise-identical re-encoding across software or hardware executions. Let x denote the resulting image in $[0,1]$, and let D_{VAE} denote the decoder's raw, unbounded output. With scaling factor $s = 0.18215$, define the floating-point codec maps used here by

$$E_s(x) = s\,\mu_{\mathrm{VAE}}(2x-1), \qquad D_s(u) = \mathrm{clip}_{[0,1]}\left(\frac{D_{\mathrm{VAE}}(u/s)+1}{2}\right). \tag{36}$$

The scaled latent law is then

$$p_{\mathrm{lat}} = (E_s)_{\#}p_{\mathrm{data}} \quad \text{on} \quad \mathbb{R}^{4\times 8\times 8} \simeq \mathbb{R}^{256}. \tag{37}$$

We train either in these coordinates or after a channelwise affine transform A. In the latter case, each of the four means and standard deviations is fit over samples and spatial positions using only the resolved training split, and the modeled target is

$$\widetilde{p}_{\mathrm{lat}} = (A \circ E_s)_{\#}p_{\mathrm{data}}, \qquad p_0 = \mathcal{N}(0, I_{256}). \tag{38}$$

Taking $A = I$ recovers the unnormalized experiment. Channel normalization is a change of latent coordinates; it is not the dimension-normalized RBF geometry of Equation (13). In particular, a raw bandwidth always refers to Euclidean distance in the actual training coordinates.

The historical WGF flow is trained entirely in its chosen $\mathbb{R}^{256}$ coordinates; D_s is used only to visualize or evaluate samples. Its actual image generator is

$$z \mapsto D_s\Big(A^{-1}(\Phi_\theta(z,0,1))\Big) \quad \text{or its multi-step counterpart from Equation (4).} \tag{39}$$

The modern LSD adaptation replaces Φ_θ by the residual map X_θ in Equation (30); the standalone latent GMD controls use their own one-step generator. All three still share the same frozen codec and its limitations. The fitted statistics are stored with the run and the standalone generator applies A^{-1} before every VAE decode. This prevents evaluation from silently interpreting normalized particles as scaled VAE latents. Image evaluation quantizes D_s to uint8 PNG after this floating-point decode. Even an exact match to the encoded data law decodes to the reconstruction law $(D_s \circ E_s)_{\#}p_{\mathrm{data}}$, not directly to p_{data}. We therefore report $D_s(E_s(x))$ as a codec reconstruction reference. It measures codec distortion but is not a formal bound on FID or KID, since a learned latent law need not equal the encoded data law.

7 Evidence on MNIST

The authoritative signal ablation contains all 36/36 planned cells: 12 variants, three seeds, 3000 updates, batch size 128, width 32, no EMA, and the same 5000 fixed latent samples per seed. Table 2 reports the final checkpoint rather than selecting the best intermediate step. Its common full recipe uses raw KL–KDE geometry, $h = 2.5$, $G = 4$, $\varepsilon = .005$, $S = 100$, $\lambda_{\mathrm{path}} = .1$, $\lambda_{\mathrm{W}} = 1$, $\lambda_{\mathrm{L}} = 1$, $\lambda_{\mathrm{E}} = 100$, and origin and arbitrary endpoint weights both equal to 5; this legacy implementation has no canonical composition loss. IS is computed with 10 splits using a project-local `SmallMnistClassifier` whose recorded test accuracy is .9831. The currently available classifier checkpoint has SHA-256 `f39ffc115dbeab5012837aaed3b3fd9c9bdfeab60cc97664b5efe0c2e560892c`, but the historical task manifests did not record that digest, so it is not a retroactive byte-level attestation of the checkpoint used by every task.

The evidence supports four narrow conclusions. First, endpoint boundaries provide most recognizable-sample quality in this intervention: removing them drops IS@1 from 6.11 to 2.70. Turning both boundary weights off also bypasses the boundary branches and changes the

Table 2: MNIST signal ablation. The IS columns are mean ± sample standard deviation over three seeds; Δ (IS@50 minus IS@1) and relative RMSE are the corresponding three-seed means.

Variant	IS@1	IS@4	IS@50	Δ	rel. RMSE
Full	$6.111 \pm .038$	$6.641 \pm .175$	$7.074 \pm .312$	$+.963$	.077
No path-WGF	$6.074 \pm .087$	$6.576 \pm .224$	$6.609 \pm .451$	$+.535$	.075
No endpoint boundaries	$2.702 \pm .346$	$2.848 \pm .367$	$2.907 \pm .332$	$+.205$	.055
No Eulerian composition	$6.174 \pm .111$	$6.390 \pm .300$	$6.028 \pm .588$	$-.145$	.119
No Lagrangian FD	$6.109 \pm .047$	$6.632 \pm .168$	$7.092 \pm .314$	$+.983$	.077
WGF signals (path + endpoint)	$6.183 \pm .107$	$6.368 \pm .304$	$5.960 \pm .685$	$-.223$	.120
Endpoint boundaries only	$6.175 \pm .042$	$6.812 \pm .157$	$6.309 \pm .301$	$+.134$	.134
Path-WGF only	$3.381 \pm .093$	$2.988 \pm .144$	$3.533 \pm .537$	$+.152$	.128
Consistency only	$1.002 \pm .003$	$1.002 \pm .003$	$1.001 \pm .002$	$-.000$	.010
No arbitrary-start boundary	$6.078 \pm .126$	$5.771 \pm .207$	$5.637 \pm .382$	$-.441$	.089
No origin boundary	$5.735 \pm .047$	$6.268 \pm .041$	$6.753 \pm .183$	$+1.018$	.089
Permuted-data full	$1.338 \pm .009$	$1.332 \pm .024$	$1.353 \pm .022$	$+.014$	.070

subsequent random-number stream, so this is not a paired counterfactual with all stochastic draws held fixed. Second, Eulerian composition is important for iteration: without it the mean NFE trend reverses. Third, retaining the two-term path-WGF block is associated with about 0.46 higher mean IS at NFE 50, but most quality remains without it. The MNIST "No path-WGF" row still retains the endpoint boundaries, Eulerian composition, and Lagrangian finite-difference loss; it is not the CelebA core in Equation (24). Fourth, this experiment resolves no benefit from Lagrangian FD: removing it changes IS@50 by +0.018, far below the seed standard deviation. Finally, removing the arbitrary-start boundary hurts MNIST at high NFE even though it helps the CelebA configuration below. Its removal is therefore a dataset-specific empirical choice, not a general property of flow maps.

These numbers do *not* demonstrate accurate PDE solution. In the full runs, tail origin/local velocity cosines are only about 0.10 and relative velocity residuals are about 1. The consistency-only negative control also shows why transition relative RMSE cannot serve as a quality metric: it attains .010 while producing one active mode and IS near 1. Conversely, the permuted-data control collapses under the normal MNIST classifier, arguing against simple evaluator or architecture leakage. We therefore describe the method as endpoint drifting with flow-map structure and WGF path matching as an imperfect auxiliary signal; CelebA below tests whether that signal remains useful in the selected core.

8 Training-disjoint local CelebA development evaluation

The historical 30,000-image package is exactly derived from aligned CelebA [12]: exhaustive decoded-pixel hashing reproduces every local PNG by cropping the official 178×218 JPEG to $(0, 20, 178, 198)$ and applying PIL bilinear resize to 64×64. It is not, however, the official training split. The local training slice 5000:30000 contains 19,994 unambiguous official-training images, 2,466 validation images, 2,530 test images, and ten resized contents whose duplicate official candidates cross partitions. Thus all historical numbers below are paired local-development results, not official CelebA test scores.

The provided 5,000-image FID directory is byte-identical to archive indices 0:5000. We train only on indices 5000:30000. A content-hash audit found one cross-slice duplicate: evaluation index 652 is byte-identical to training index 20010. We therefore exclude index 652 and use the remaining 4,999 images as the canonical no-exact-train-overlap development reference. Geometry and bandwidth calibration use only the 25,000-image training pool. Identity overlap remains possible and is also present in the canonical image-level CelebA partitioning.

For future confirmation, we have separately materialized the official training/validation/test

partitions with 162,770/19,867/19,962 images under `data/celeba64_official_bilinear`. Its manifest SHA-256 (prefix `27c3098d`, suffix `5048482`) is recorded in full in the dataset audit and binds the source archive, partition file, identities, exact transform, and per-image records. No FID/KID value reported in the tables comes from this official partition, and no score has been read from its test split. A dedicated fail-closed MAE wrapper and a pre-result 192-epoch protocol enforce training-only access and rehash all 162,770 PNGs before and after preload. The MAE completed exactly 121,920 updates in two allocations totaling 5:39:02, and its final EMA feature checkpoint has SHA-256 prefix `9d088740`.

The downstream direct-RGB path is also frozen in a dedicated wrapper and pre-result protocol. It requires the exact EMA MAE at update 121,920 with an externally supplied artifact hash and training-signature hash, segments one fresh generator trajectory at 25k/50k/75k/100k, and permits selection only on the 19,867-image official validation partition. The 19,962-image official test partition is sealed and absent from the launcher. Its 56 dedicated and non-regression tests pass. The first 0–25k segment completed without restart in 4:12:41 of trainer time and 4:22:20 of Slurm wall time; its periodic 25k checkpoint has SHA-256 prefix `64c9c324`. The terminal postflight records `fid_computed=false` and `test_read=false`. After an independent terminal audit, the exactly pinned 25k–50k continuation was submitted as job `759647`; it is pending cluster resources. An official validation result therefore does not yet exist.

8.1 Historical WGF confirmation and retained controls

For the historical WGF study, the objective, support geometry, and inference NFE were selected using this same training-disjoint set. It is therefore a development evaluation set, not an untouched final test set. The source-locked confirmation deliberately reuses that same development set; it improves experimental control and provenance, but does not create a new independent evaluation. Every replicated row below uses final EMA weights, 4,999 generated images, the same sample-index-seeded standard-normal draws, the same real features, and training seeds $41, 42, 43$. We aggregate only the common NFE grid with complete three-seed coverage, $1, 4, 16$. KID spread in the tables is the sample standard deviation across training seeds, not the much smaller subsampling error internal to a single KID evaluation.

The main table comes from this new confirmation, for which deterministic algorithms were enabled and exact source bytes were archived for every training and evaluation task. The historical replicated runs used in the later ablation tables were seeded but recorded `deterministic=false`; their values are retained as historical evidence rather than mixed into the confirmation.

The historical 1,332,036-parameter latent U-Net is trained for 5,000 updates on one node with eight MI210 GPUs (eight ranks), with local batch 128 and global batch 1024. We use raw latent geometry, $h = 3$, $G = 4$ synchronized time groups, hence 1024 data supports and 256 model supports per time, $\varepsilon = .005$, $S = 100$, $\eta = 1$, and EMA decay .999 after 500 updates. AdamW uses learning rate 2×10^{-4}, betas $(.9, .999)$, zero weight decay, and gradient clipping at 1. The inference partition is uniform. KID uses 100 repetitions of subsets of 1,000 images with seed zero. We use the standard FID [10] and KID [11] definitions. FID, KID, and the feature trace use the 2,048-dimensional pool3 features from `pytorch-fid` 0.3.0 [13]; the Inception weights have SHA-256 `6726825d0af5f729cebd5821db510b11b1cfad8faad88a03f1befd49fb9129b2`. Feature trace means the trace of the empirical covariance of those features. All single-seed screens and historical controls are labeled as such rather than pooled with the replicated results.

For every historical core seed, four transitions improve on one: the paired FID change is $-15.793 \pm .927$. From four to sixteen it is $+.338 \pm .838$, so the method plateaus rather than converging monotonically with NFE. For B the corresponding changes are -19.299 ± 2.449 and $+1.347 \pm 1.023$. The paired B-minus-C FID is $+3.649 \pm 1.034$, $+.144 \pm 3.805$, and $+1.152 \pm 2.413$ at NFE 1, 4, and 16, respectively; at NFE four its three seedwise signs are negative, positive, and negative. Thus C is both the simpler objective and the one with the lowest confirmed mean FID at NFE four, but three seeds do not resolve a population advantage between C and the

Table 3: Historical WGF-core, source-locked CelebA-64 confirmation from final EMA weights. Values are mean ± sample standard deviation over three training seeds. Feature trace is reported without a direction arrow; its value on the common 4,999-image real set is 126.959.

Method	NFE	FID↓	KID×10^3 ↓	feature trace
Core C	1	$61.556 \pm .590$	$52.714 \pm .374$	$130.603 \pm .429$
Core C	4	45.762 ± 1.513	33.716 ± 1.306	114.241 ± 2.271
Core C	16	$46.101 \pm .685$	$33.143 \pm .807$	106.018 ± 3.613
Eulerian-regularized B	1	$65.205 \pm .872$	57.601 ± 1.102	136.614 ± 1.106
Eulerian-regularized B	4	45.906 ± 2.293	34.691 ± 2.142	119.408 ± 5.080
Eulerian-regularized B	16	47.253 ± 1.759	36.542 ± 2.365	117.054 ± 4.757
Historical optimizer-time KDE drift	1	66.020 ± 1.002	56.241 ± 1.383	$129.398 \pm .921$

Eulerian-regularized variant. Both variants fall below the real-set feature trace at higher NFE, with stronger contraction for C. The optimizer-time drift result is a separately tuned reference, discussed below, rather than a matched ablation.

Table 4: Contextual references and controls. These are fixed-distribution references or single runs, not three-seed comparisons with Table 3.

Reference or control	FID↓	KID×10^3 ↓	feature trace
Real–real, 2,499 vs. 2,500 disjoint images	10.996	.133	127.112
VAE reconstruction reference	15.814	10.822	125.172
$D(z),\ z \sim \mathcal{N}(0, I)$	352.095	455.074	117.792
Channel-moment Gaussian	324.620	399.839	140.473
Diagonal latent Gaussian	282.499	341.550	157.866
Full-covariance latent Gaussian	79.481	72.512	147.396
Historical raw-pixel KDE/WGF screen, one seed	177.979	199.277	111.260

For the real–real row, the displayed trace is that of the 2,500-image comparison half; the 2,499-image reference half has trace 126.775. Neither is the trace 126.959 of the common 4,999-image real reference used in the main evaluation. All three Gaussian controls are fitted only on training indices 5000:30000.

The Gaussian controls diagnose the latent problem. The channel control fits a mean and variance per channel, and the diagonal control fits all 256 coordinate means and variances; both remain poor. The decrease from FID 282.499 for the diagonal Gaussian to 79.481 for the full-covariance Gaussian is consistent with a large role for cross-coordinate correlations, although changing the entire covariance family is not a causal isolation of that mechanism. The historical learned WGF flow reaches below the dense Gaussian at one NFE and improves substantially when composed. Thus second moments capture a substantial part of the measurable gap without isolating its cause, and they do not capture all of the target structure. The VAE reconstruction row is only a codec reference and is not a bound on a learned generator's FID.

The three-seed optimizer-time drift confirmation uses raw bandwidth $h = 4$ and drifting step 1, whereas the learned-flow rows use $h = 3$; it is therefore a separately tuned reference rather than a one-factor ablation of the flow map.

An exploratory seed-42 screen also tests whether a train-fitted dense Gaussian prior or a residual endpoint parameterization improves the selected B-style objective. Table 5 reports all 4,999-image evaluations. The standard/absolute row is a historical control produced under an earlier source state; it is not contemporaneous with the other three rows, so the table is descriptive rather than a causal 2×2 ablation.

The absolute form in Equation (6) has an explicit source coefficient $(1 - b)/(1 - a)$ that telescopes and vanishes at the terminal time. The residual alternative, $\Phi(x, a, b) = x +$



Figure 1: Fixed qualitative grid from historical WGF confirmation seed 42. Columns are image indices 0–7, selected in advance with no image-quality filtering. All generated rows use the same index-seeded prior draw within a column; rows show historical optimizer-time KDE drift at NFE one and WGF variants C and B at NFE 1, 4, and 16. The real row is an unpaired reference and is not the target corresponding to any generated column. Thus neither the real images nor the generated images were selected to make a favorable visual comparison.

$\alpha(a,b)r_\theta(x,a,b)$, retains an explicit unit skip at every call; it can in principle learn a cocycle, but it has no architectural source-skip cancellation. Its severe degradation under composition with the standard prior is therefore consistent with the architectural argument. The dense prior stabilizes that residual variant but does not beat the simpler standard-prior/absolute control at NFE four. Given the single seed and historical-control confound, this screen motivates keeping the simpler design rather than establishing a population effect.

Moving from $3 \times 64 \times 64 = 12{,}288$ pixels to $4 \times 8 \times 8 = 256$ latents reduces dimension by a factor of 48 and delegates local image reconstruction to the frozen decoder. In raw pixels, Euclidean kernel distances mix alignment, illumination, texture, and local detail; the historical raw-coordinate KDE/WGF screen remains at FID 177.979. The newer direct-RGB GMD control instead reaches FID 53.262 with learned MAE features, and its matched MAE+ConvNeXt branch reaches 18.887, showing that raw pixels are not intrinsically the failure mode and causally isolating a large feature effect within the RGB recipe. Comparisons with the latent and historical KDE models still change architecture, feature geometry, objective, and training settings, so they

Table 5: Exploratory prior and endpoint-parameterization screen (one seed). The entries are FID at the indicated NFE.

Prior	endpoint parameterization	NFE 1	NFE 4	NFE 16
Standard (historical)	absolute	64.618	44.869	46.748
Standard	residual	72.471	320.822	268.365
Full train Gaussian	absolute	56.066	49.012	50.136
Full train Gaussian	residual	53.572	63.524	57.897

do not causally isolate dimension.

Table 6: Historical replicated NFE-four objective and support comparison. These paired values are kept together under their original historical protocol and are not replaced by the later confirmation values in Table 3.

Configuration	FID↓	KID$\times 10^3$ ↓
A: origin + WGF + Eulerian, no canonical composition	44.312 ± 1.331	31.65 ± 1.49
B: A + canonical composition	$44.014 \pm .884$	32.34 ± 1.58
C: origin + WGF + canonical composition, no Eulerian	45.761 ± 1.511	33.72 ± 1.31
B-style objective with $G = 2$	46.326 ± 1.479	35.67 ± 1.29
B-style objective with $G = 2$, no canonical composition	69.591 ± 38.550	62.66 ± 47.82
Origin and arbitrary endpoint boundaries + WGF + Eulerian, no canonical	49.286 ± 2.620	38.39 ± 2.20

Because these rows were trained together under their historical protocol, the table supports a deliberately narrow matched mechanism claim. Canonical and Eulerian composition are partially substitutable: either produces a useful multi-step model when combined with endpoint drift and the WGF block. Doubling the model-reference support per time by changing $G = 4$ to $G = 2$ does not help; without the canonical term, one of three $G = 2$ seeds collapses to FID 114.095. Thus support size and temporal coverage cannot be reduced to a monotone "more particles is better" rule. Removing the arbitrary-start boundary improves NFE-four FID relative to the matched two-boundary runs in all three seeds (49.286 to 44.312 in the mean), while degrading the direct-map mean from 63.876 to 68.191. A plausible explanation is that its detached terminal update acts through a different parameter-gradient path and can compete with the origin endpoint update before composition is accurate; we did not measure per-loss gradient cosines, so this remains an interpretation.

A seed-42-only factorial screen further checks the three core mechanisms. At NFE four, C gives FID 44.028; removing its WGF block gives 56.404, removing canonical composition gives 78.856, and removing both gives 78.175. Replacing canonical composition by Eulerian composition gives 42.792; keeping both consistency terms but removing WGF gives 48.357. Removing the origin endpoint update from WGF plus both consistency terms fails at 260.195. These are exploratory point estimates, and the last intervention also changes the executed boundary branch and subsequent random-number stream.

Tail diagnostics keep the theoretical claim proportional to the estimator. They are means over the final 10% of optimizer-time records within each run, then mean ± sample standard deviation over seeds; they describe the online raw network, not the final EMA used for image evaluation. KDE ESS is estimated from one query in each of four time groups on each of eight ranks at every recorded step, then averaged across ranks; velocity RMS and residual ratios use the complete batch. For C, attraction/repulsion ESS are 129.554 ± 1.992 and 48.851 ± 2.130, velocity RMS is $.024358 \pm .000215$, and origin/local relative WGF residuals are $1.0408 \pm .0313$ and $1.1364 \pm .0657$. For B, attraction/repulsion ESS are 128.310 ± 1.154 and 40.986 ± 1.478, velocity RMS is $.0281545 \pm .000597$, and the two residuals are $.928874 \pm .01036$ and $.995754 \pm .04413$. Separately, within each run the final-EMA transition relative RMSE averages the 10 nontrivial time pairs on the grid $\{0, .25, .5, .75, 1\}$, evaluated on 16 fixed prior samples; the reported

three-seed mean ± sample standard deviation is $.183880 \pm .022878$ for C and $.178625 \pm .018277$ for B. Residuals near one support the language of an imperfect directional teacher, not an accurately solved WGF PDE.

8.2 Observed single-seed development controls

Table 7 keeps the later methods separate from the historical three-seed confirmation. Every entry is one training seed (42), 4,999 generations with evaluation seed 42001, and the same reused training-disjoint development reference. The entries are point estimates; no across-seed uncertainty is available.

Table 7: Observed CelebA-64 single-seed development measurements. "Update" identifies the evaluated training state and is not a matched compute budget across methods. KID is multiplied by 10^3.

Method	update	weights	NFE	FID↓	KID×10^3 ↓
Pure latent LSD, LR .003	100k	raw	1	34.377	26.487
Pure latent LSD, LR .003	100k	raw	2	32.757	25.335
Pure latent LSD, LR .003	100k	raw	4	31.716	25.177
Pure latent LSD, LR .003	100k	raw	8	31.119	24.843
Pure latent LSD, LR .003	100k	raw	16	30.799	24.595
Pure latent LSD, LR .003	100k	EMA	1	32.358	24.625
Pure latent LSD, LR .003	100k	EMA	2	29.740	22.314
Pure latent LSD, LR .003	100k	EMA	4	28.261	21.141
Pure latent LSD, LR .003	100k	EMA	8	27.637	20.670
Pure latent LSD, LR .003	100k	EMA	16	27.613	20.792
Latent GMD-v1	30k	EMA	1	37.413	28.333
Latent GMD-v2, MAE-v2 `official_70`	30k	EMA	1	39.379	30.318
Direct-RGB GMD	10k	EMA	1	53.262	44.640
Direct-RGB GMD + ConvNeXt-V2	10k	EMA	1	18.887	8.841

The pure LSD run uses a 21,006,724-parameter latent U-Net, global batch 256, RAdam, and the predeclared 100k checkpoint as its terminal result. Both raw and EMA FID decrease over the five reported NFEs, but NFE sixteen costs exactly sixteen network calls per image and the trend has not been replicated across training seeds. The diagnostic 75k EMA checkpoint reaches FID 27.234 at NFE sixteen, slightly below the terminal 27.613; it cannot replace the registered 100k endpoint and shows that optimizer-time progress is not monotone. Across the table, feature encoders, architectures, batches, optimizer settings, and budgets differ. Consequently these rows support engineering comparisons on this development split, not causal rankings or a population claim.

The exception to the general non-comparability of rows is the last two: their pre-registered design changes only the addition of the frozen 13-family ConvNeXt-V2 extractor. Their 53.262-to-18.887 difference is therefore a causal feature-stack effect for this seed and recipe. It remains a development screen rather than an official-test or multi-seed performance estimate.

Table 8 is the causal comparison that the standalone rows cannot provide: pure LSD, D2, and D4 resume the same raw 30k state with matched RAdam, data-stream, and random-number states. D2 and D4 differ from pure only by the ramped GMD terminal anchor during updates 30k–40k; D4's target coefficient is exactly one half of D2's.

On the registered raw contrast, D2 improves NFE-one FID by 11.221% and NFE-one KID by 16.178%. At NFE sixteen it also improves, rather than merely preserving, FID by 3.490% and KID by 6.251%; its raw FID is nonincreasing over the entire NFE grid. Thus all pre-specified gates pass and the pilot is declared promising. This is stronger evidence than a post-hoc best-checkpoint comparison, but it remains one training seed on a reused development split and

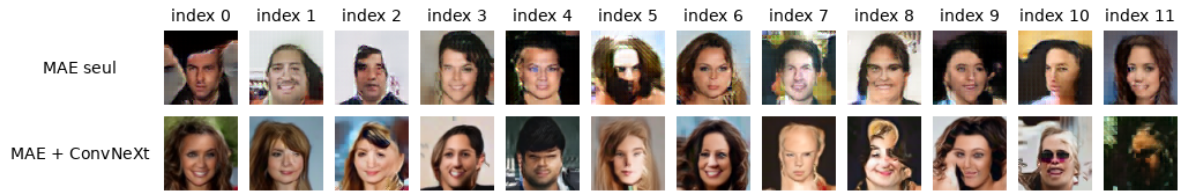


Figure 2: Fixed-index qualitative companion to the direct-RGB causal screen. The upper row uses the 82 MAE families alone and the lower row adds the 13 ConvNeXt-V2 families. Columns use identical generation indices and seeds 42,001–42,012, fixed before visual inspection; no image was ranked or quality-selected. Both rows are EMA, update 10k, NFE one, and retain native 64×64 pixels.

Table 8: Pre-registered paired CelebA-64 pilots at 40k. Each body entry is FID / KID$\times 10^3$ on the same 4,999-image development reference. Raw was primary for D2; EMA was primary for the later D4 half-dose ablation.

NFE	pure raw	D2 raw	D4 raw	pure EMA	D2 EMA	D4 EMA
1	37.625/30.671	33.403/25.709	34.328/27.213	34.130/27.267	32.693/25.634	33.308/26.360
2	34.263/28.080	31.462/24.263	32.665/25.782	29.951/23.104	29.211/22.194	29.345/22.358
4	32.520/26.676	30.769/24.146	32.164/25.610	28.388/21.676	28.154/21.361	28.153/21.428
8	31.725/25.944	30.532/24.183	31.797/25.534	27.862/21.218	27.941/21.309	27.810/21.159
16	31.503/25.801	30.404/24.188	31.389/25.213	27.731/21.083	27.868/21.308	27.737/21.097

does not establish a population or method-wide advantage for GMD-regularized LSD.

D4 also passes every registered raw check: its raw FID decreases monotonically from 34.328 to 31.389, and at NFE sixteen it remains 0.364% below pure LSD. On its pre-registered primary EMA contrast it improves FID by 2.409% and KID by 3.327% at NFE one, and it is still slightly better at NFE eight. At NFE sixteen, however, FID is 27.7370891399 versus 27.7305457830 (+0.0065433568, or +0.0236%), while KID is 0.0210968890 versus 0.0210828191 ($+1.406998 \times 10^{-5}$). Both inequalities were strict and all gates were conjunctive, so the locked verdict is `NO_GO`. This is a precise negative result for the half-dose rule: it nearly removes D2's large-NFE EMA penalty, but does not remove it completely. It is not evidence of a practically large degradation, nor a license to tune the coefficient after seeing these metrics.

9 Limitations

The historical WGF method depends on a batch score estimator whose quality degrades in high dimension and in low-density regions. Global DDP support improves particle count but does not remove KDE bias. Endpoint semi-gradient updates can produce useful samples even when the learned local field poorly matches the nominal WGF. That method therefore should not be called an accurate WGF solver unless residual and distributional evidence improve substantially. The modern LSD objective does not use this KDE estimator, and the GMD adaptations instead depend on the geometry of frozen learned features. A frozen VAE changes the modeled target and introduces codec distortion, which the reconstruction reference measures without constituting a metric bound.

The training-disjoint set was reused adaptively to choose losses, support, checkpoints, and NFE, and it is not an untouched test set. Only the historical WGF C/B variants and historical optimizer-time KDE drift have three confirmatory training seeds. Pure LSD, D2, D4, latent GMD-v1, latent GMD-v2, both direct-RGB GMD branches, pixel training, Gaussian controls, and several factorial cells remain single-seed development measurements or fixed-distribution



Figure 3: Paired qualitative companion to Table 8. Columns are the first ten generation indices, fixed before visual inspection; each column therefore reuses the same indexed CPU noise across pure LSD and D2. Rows show raw and EMA weights at NFE one and sixteen. Tiles retain their native 64×64 pixels and no image was ranked or quality-selected. The grid is descriptive rather than a statistical comparison; its JSON sidecar records the row order and SHA-256 of every selected PNG and source manifest.

references. The RGB pairing identifies the effect of adding ConvNeXt for that exact seed, but not its population effect. In particular, the small historical B–C difference is unresolved, and neither the successful D2 pilot, the narrowly negative D4 result, nor the other newer point estimates can establish a population ordering.

NFE four is the historical WGF operating point, not the beginning of a monotone convergence curve: at NFE sixteen its FID plateaus and feature covariance can contract, especially for C. The pure-LSD 100k, D2 40k, and D4 40k raw FID curves decrease over the observed NFE grid through sixteen, but each curve is one seed and none justifies extrapolation. Moreover, the diagnostic pure-LSD 75k EMA endpoint is slightly better than 100k at NFE sixteen, so training-update progress is not monotone. The historical canonical composition loss is sampled only on model-reached states and time pairs, not enforced globally. The frozen VAE is a strong external inductive bias, while comparisons between latent and direct-RGB GMD change both architecture and feature geometry. All results are limited to CelebA-64 and this reused development protocol. The historical WGF samples at FID near 46 are recognizable as faces but still show warped geometry, artificial colors, and merged features; the number should not be read as photorealism. We provide no likelihood, state-of-the-art comparison, or evidence of transfer to another high-dimensional dataset.

The canonical latent cache predates byte-level VAE attestation: it records the local model path, identifier, scaling factor, and encoding rule, but not hashes of the encoder weights used

when the cache was built. We therefore re-encoded the same ordered 30,000-file slice twice with the current float32, batch-128 pipeline on the same MI210 node. All three caches have the same core metadata and ordered source manifest, but none of their latent tensors are exactly equal. For historical–attested, historical–repeat, and attested–repeat, respectively, the changed-value counts out of 7,680,000 are 6,108,402, 6,107,667, and 6,110,122; the MAEs are $1.11973642401\times10^{-7}$, $1.11860632774\times10^{-7}$, and $1.12148330792\times10^{-7}$; the RMSEs are $1.6602572978\times10^{-7}$, $1.65839475814\times10^{-7}$, and $1.67202344252\times10^{-7}$; and the maximum absolute differences are $1.45733356476\times10^{-5}$, $1.72760337591\times10^{-5}$, and $1.46757811308\times10^{-5}$. Thus the historical–current discrepancy has the same observed scale as this single current–current repeat control. This is limited compatibility evidence, not a claim of identity, interchangeability, equivalent caches, or invariant downstream training, generation, or metrics; no such downstream invariance experiment was run, and the historical encoder-weight bytes remain unattested.

All nine final confirmation checkpoints record completed step 5,000, `deterministic=true`, canonical-cache SHA-256
`a7e5daf36e5ad2fc45eb4972b2e394b519438143892509073f24b29f5fdc2ebd`,
and selected-training-values SHA-256
`4ad88bd829829279d807db7da240e76569e144474645a2bd9c413ef870ccd83a`.

This binds the confirmation to the exact canonical cache and resolved training tensor used; it does not retroactively attest the encoder weights that produced that cache or establish equivalence to either re-encoding.

The two current caches record the same 18-file recursive VAE inventory, which matches the directory at audit time. That scope is operationally fragile: it includes 13 mutable `.cache` entries, five of them lock files. Future attestation should hash the stable configuration and inference-weight artifacts, record which weight file the loader selected, and exclude cache, lock, download, README, and version-control metadata. This recommendation was not applied retroactively to the reported caches.

The cache-generation jobs did not embed source-file hashes; the code hashes in the audit report are therefore audit-time evidence, not execution-time source snapshots.

The historical CelebA training provenance records per-file source hashes but does not contain retrievable byte snapshots; several key files were untracked, so those hashes alone cannot reconstruct the exact source. The MNIST task manifests are weaker still: they archive neither code hashes nor the classifier checkpoint hash. The current trainer and confirmatory launcher are configured to archive the exact source bytes and a hashed inventory; the completed confirmation contains those snapshots for all nine training tasks and all 21 evaluation cells. This does not repair the historical artifacts.

The 54 generation manifests behind the six historical replicated CelebA groups contain a legacy field claiming batch-size invariance. That field is erroneous for the neural transition and decoder and must be disregarded; only the rowwise construction of the prior samples is partition-invariant. Those historical artifacts were generated with batch size 256, a float32 decoder, and an AMD Instinct MI210. We make no claim that another generation batch would reproduce them bitwise. Confirmation manifests state these scopes separately.

Channel normalization is also not an invariance claim. Retaining $p_0 = \mathcal{N}(0, I)$ after changing target coordinates changes the implied source law in raw latent coordinates. The selected runs use $A = I$.

10 Conclusion

The historical WGF study shows that endpoint drift, a finite-particle directional signal, and learned composition can coexist in one two-time map. Its source-locked three-seed result identifies Core C as the simpler of two closely performing WGF objectives at the NFE-four operating

point, without resolving a population advantage over B. That evidence remains useful, but it does not describe the newer objective.

At the predeclared pure-LSD 100k terminal, raw FID is 34.377 at one network evaluation and 30.799 at sixteen; EMA gives 32.358 and 27.613. Pure LSD contains no drifting/GMD loss. The paired D2 pilot does add a weak terminal GMD anchor and passes every registered 40k gate: relative to pure raw, it improves one-evaluation FID from 37.625 to 33.403 while also improving the sixteen-evaluation result from 31.503 to 30.404. This supplies a positive answer for the exact pilot, not a general superiority result: D2 is single-seed, roughly twice as slow per observed update, and is not the published long–short method. The pre-registered D4 half-dose retains the raw gain and improves EMA at one and eight evaluations, but misses its two strict EMA NFE-sixteen gates by 0.00654 FID and 1.41×10^{-5} KID; its formal verdict is therefore `NO_GO`. This narrows the useful dose region without justifying post-hoc retuning. The standalone latent and direct-RGB GMD values are likewise single-seed controls of adapted CelebA recipes. In particular, augmenting the local direct-RGB MAE features with ConvNeXt changes FID from 53.262 to 18.887 under a one-factor pre-registered contrast, while making each update 4.07× slower. This establishes feature geometry as a major local bottleneck and directly refutes an intrinsic inability to train GMD without a VAE on these CelebA pixels; it does not establish an official-test or population result. Thus "drifting as a flow map" is a descriptive organizing theme, already related theoretically to long–short flow maps by prior work, rather than a novelty or broad superiority claim. Multi-seed confirmation, an untouched evaluation split, and transfer remain open.

A Reporting checklist

Every final run should identify: exact source snapshots and their hashes (a revision and dirty-diff hash alone are insufficient for untracked files); dataset file list and split indices; VAE identifier, scaling factor, posterior-mean (sampling-free) encoding rule, encoder/decoder file hashes, and any fitted latent normalization; architecture and parameter count; optimizer, update count, local/global batch, ranks, and wall-clock time; method identity and all active objective coefficients. Historical WGF runs must additionally record h, RBF geometry, S, ε, η, and global-support/synchronized-time flags. LSD runs must record the interpolant coupling, diagonal/off-diagonal split, residual parameterization, time-JVP, learned log-variance policy, and whether the stopped target is raw or EMA. GMD runs must record the feature encoder and weight kind, feature families, temperatures, positive/negative supports and banks, and conditioning mode. Every method must record EMA policy, raw/EMA evaluation choice, checkpoint selection rule, sampling partition and NFE, latent seed manifest, evaluator version, and reference-feature hash. This metadata is part of the result, not an implementation detail.

B CelebA source tables

The source-locked confirmation, its training summary, and the qualitative-grid sidecar are stored at

```
runs/celeba_latent_wgf/latent_final_confirmation_v12/protocol_lock.json
runs/celeba_latent_wgf/latent_final_confirmation_v12/evaluation/final_results.json
runs/celeba_latent_wgf/latent_final_confirmation_v12/evaluation/final_results.md
runs/celeba_latent_wgf/latent_final_confirmation_v12/training_summary.json
runs/celeba_latent_wgf/latent_final_confirmation_v12/training_summary.md
runs/celeba_latent_wgf/latent_final_confirmation_v12/evaluation/protocol_lock.json
latex/figures/celeba_final_seed42_nfe_grid.png
latex/figures/celeba_final_seed42_nfe_grid.json
```

The final aggregation validates all 21 expected evaluation cells, including their training checkpoints, generation and evaluator provenance, archived source snapshots, and protocol locks. The grid JSON records its exact row and column order, fixed indices, input hashes, and output hash.

The historical aggregate, prior-screen, and cache-audit artifacts are

```
runs/celeba_fid/summary.json
runs/celeba_fid/paper_results.json
runs/celeba_fid/paper_results.md
runs/celeba_latent_wgf/latent_prior_endpoint_screen_v11/report.json
runs/celeba_latent_wgf/latent_prior_endpoint_screen_v11/report.md
runs/celeba_latent_wgf/latent_cache_attestation_v12/report.json
runs/celeba_latent_wgf/latent_cache_attestation_v12/report.md
```

The summary snapshot validates every admitted evaluation JSON and records invalid legacy cells. The paper-result JSON adds protocol hashes, per-seed values, aggregates, missing cells, and paired NFE deltas; its Markdown rendering is the compact human-readable view. The cache-attestation JSON is the machine-readable comparison of the historical cache and two current re-encodings; its Markdown rendering records the audit, inventory-scope caveat, and reproduction command. The confirmation table and final diagnostics were checked against the final-result and training-summary JSON files above; the historical ablation tables remain checked against their historical snapshots. The historical diagnostic sources retained for those comparisons are

```
runs/celeba_latent_wgf/latent_semigroup_v3/training_summary.json
runs/celeba_latent_wgf/latent_boundary_semigroup_replication_v8/training_summary.json
runs/celeba_latent_wgf/latent_objective_ablation_v7/training_summary.json
runs/celeba_latent_wgf/latent_semigroup_no_euler_replication_v10/training_summary.json
```

B.1 Modern LSD/GMD development artifacts

The observed rows in Table 7 are anchored by the following human-readable artifacts (their adjacent JSON/manifests contain the full machine-readable provenance):

```
runs/celeba_latent_lsd/lsd_independent_std1_base256_seed42_lr003_screen10k_v1/
  evaluation/lr003_continued100k_steps50k75k100k_raw_ema_nfe124816_
  4999_seed42001_sha256_serialized_v1/aggregate_strict/summary.md
runs/celeba_official_drifting/evaluation/vae_float32_v2/metrics/
  mae_gmd_v1__step_00030000.md
runs/celeba_official_drifting/evaluation/
  mae_gmd_v2_feature30k_official70_base256_seed42_30k_v1_curve_ema_4999_
  sha256_serialized_cache_v2/aggregate/summary.md
runs/celeba_pixel_gmd/generator/
  direct_rgb_gmd_screen_ditb_prodmae_seed42_10k_v1/
  evaluation/direct_rgb_gmd_screen_4999_seed42001_v1_retry1/
  aggregate_strict_v1/terminal_contrast.md
runs/celeba_pixel_gmd/generator/
  direct_rgb_gmd_screen_ditb_prodmae_nondec_convnextbase_fp32_seed42_10k_v1/
  evaluation/convnext_terminal_ema10k_nfe1_4999_seed42001_v1/
  aggregate_strict_v1/summary.json
CELEBA_PIXEL_GMD_CONVNEXT_SCREEN_EVALUATION_PROTOCOL_V1.md
CELEBA_PIXEL_GMD_CONVNEXT_OFFICIAL_CAUSAL_CONFIRMATION_PREPROTOCOL_V1.md
CELEBA_PIXEL_GMD_FIDELITY_GATE_RESULT.json
runs/celeba_diagnostics/official_gmd_kernel/
OFFICIAL_GMD_KERNEL_DIAGNOSTIC.md
CELEBA_PIXEL_GMD_MAE_OFFICIAL_SPLIT_PROTOCOL.md
CELEBA_PIXEL_GMD_GENERATOR_OFFICIAL_SPLIT_PROTOCOL.md
runs/celeba_latent_lsd/lsd_independent_std1_base256_seed42_lr003_
  ramped_gmd10pct_40k_v1/evaluation/ramped_anchor40k_vs_pure40k_
  raw_ema_nfe124816_4999_seed42001_sha256_serialized_miopen_v2/
```

aggregate/aggregate.md
CELEBA_LSD_GMD_D4_HALF_ANCHOR_PROTOCOL.md
CELEBA_LSD_GMD_D4_HALF_ANCHOR_40K_AGGREGATION_RECOVERY_V12.md
CELEBA_LSD_GMD_D4_HALF_ANCHOR_40K_AGGREGATION_AUTHORIZATION_V12.json
latex/figures/celeba_d2_paired_indices0_9.png
latex/figures/celeba_d2_paired_indices0_9.json
latex/figures/celeba_rgb_gmd_convnext_causal_grid.png
latex/figures/celeba_rgb_gmd_convnext_causal_grid.json

The strict pure-LSD 100k aggregate JSON and Markdown have file SHA-256
ffcc6a3bd9c200e71d232ffb031dd806289771d4180274b6f32aea5e4ca03b81
and
fbdbd98ddeb08bcb04d7ebe7174c81952c23ef655ecfa4504bfb4c764e3ce289,
respectively. Its embedded summary hash is
64f5bea896d427a7441d8c76466ce729dd731c799b0cfec14c23eb847e0135df,
and its canonical audit hash is
b51389fb1cbeecfd74e02d19d484a80c43c89923ac81b177e57a4775f87c98ab.
The immutable D2 aggregate JSON and Markdown have file SHA-256
e4045915146e3673637d2e92bb56ea4af75cf75373113947a26630a4faf18e4b
and
dd775723fc1da95f0f6d0a03c60115a50bc58fd7b86bb8ce048029e2d2decac2,
respectively; it validates all 20 cells and records `promising=true`. The D4 terminal checkpoint has SHA-256
eae7c835b8d65ceb2bdebdea1e0b0c71f0a2120ed221056a3dd2de507aedce37.
Its pre-result protocol and V12 read-only authorization have SHA-256
7054baa5e1effe087dc2ae82fb8bdfb57244002be2bc2f41587eba969e6b05b0
and
f6d86df24d76424ab0cc2192e65f9a6e87fd09a21f97506670ff81e0f5d9042c,
respectively. The terminal V12 audit returned successfully without writing a result file, revalidated all ten D4 metric cells, and recorded recomputation trace SHA-256
a703ff0ef6f13fbd932f6045c73f7af984134d6e2faf00059676bf0d773e61ec;
its locked decision is `NO_GO`. The fixed-index D2 grid PNG and provenance sidecar have file SHA-256
7c10d0f84fba6828d03a0423a27ef9a8dbe94a1c5e0364e4b8aff1cc01f2a56f
and
b5b9da3d2d0364d03e3fcfcb480251cc7d9a623a9af3bbaed9a8d367f33ccff5,
respectively. The GMD-v2 aggregate declares logical SHA-256
06de7cfcdb6675a7f7013af1a61dd01f5b7b7acc7d984463fd1eecbb13361e7e.
The direct-RGB gate attestation has file SHA-256
e4d1a3e9dbf3402142f271d878591f91aa33c1a1d1818313ebee6eef86ad58c7.
The terminal ConvNeXt summary has file SHA-256
73127f80e2fc0f2e2abf317e014d6a88166d382b398a886678a8662eec103b91,
and the fixed-index RGB qualitative grid and its JSON sidecar have file SHA-256
68ed3b3ad1c6bc73a38ab2faac3f706ac1c1da6e8b2ec2342163642f25671a85
and
ec6e8f6dddc5dde8837d51ff83a4b1b3fa779abc6db91b19e295c1bd15911b65,
respectively. The GMD-v1 30k metric Markdown has file SHA-256
e552ac58201823ce8b4773f156ad902e387b5fdf464a5e4a84a6880ea87914e1.
The loss-port parity checks are

tests/test_official_drifting_jax_golden.py
tests/test_official_drifting_ddp_parity.py

with file SHA-256 `0e72f34179adf7604c83d227df2bdcaa08bb69eff0239b26fc84a8ae4f37d38b` and `f6f1d51a794589dd6cda104dabbc1d478ce4090afa9f8dfbb4e3d0bffc590c76`, respectively. The D2 training and evaluation documents are

`CELEBA_LSD_GMD_RAMPED_ANCHOR_PROTOCOL.md`
`CELEBA_LSD_GMD_RAMPED_ANCHOR_40K_EVALUATION_PROTOCOL.md`

They are the pre-result protocol locks; the immutable aggregate above is the separate observed D2 result.

C MNIST source table

The values in Table 2 come from the experiment aggregation at

`runs/mnist_wgf/signal_ablation_v1/report.md`

with detailed cells in `runs.csv` and `summary_by_variant.csv`. The older exploratory numbers retained in `latex/main_exp.tex` are historical and must not be mixed with this controlled ablation.

References

[1] E. Parzen. On estimation of a probability density function and mode. *The Annals of Mathematical Statistics*, 33(3):1065–1076, 1962. https://doi.org/10.1214/aoms/1177704472.

[2] R. Jordan, D. Kinderlehrer, and F. Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998. https://doi.org/10.1137/S0036141096303359.

[3] Q. Liu and D. Wang. Stein variational gradient descent: A general purpose Bayesian inference algorithm. In *Advances in Neural Information Processing Systems 29*, 2016. https://papers.neurips.cc/paper/2016/hash/b3ba8f1bee1238a2f37603d90b58898d-Abstract.html.

[4] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. https://openreview.net/forum?id=PqvMRDCJT9t.

[5] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever. Consistency models. In *Proceedings of the 40th International Conference on Machine Learning*, pages 32211–32252, 2023. https://proceedings.mlr.press/v202/song23a.html.

[6] N. M. Boffi, M. S. Albergo, and E. Vanden-Eijnden. How to build a consistency model: Learning flow maps via self-distillation. *arXiv preprint arXiv:2505.18825*, 2025. https://arxiv.org/abs/2505.18825.

[7] M. Deng, H. Li, T. Li, Y. Du, and K. He. Generative modeling via drifting. *arXiv preprint arXiv:2602.04770*, 2026. https://arxiv.org/abs/2602.04770.

[8] Z. Li and B. Zhu. A long–short flow-map perspective for drifting models. *arXiv preprint arXiv:2602.20463*, 2026. https://arxiv.org/abs/2602.20463.

[9] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022. https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.

[10] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems 30*, 2017. https://proceedings.neurips.cc/paper/2017/hash/8a1d694707eb0fefe65871369074926d-Abstract.html.

[11] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton. Demystifying MMD GANs. In *International Conference on Learning Representations*, 2018. https://openreview.net/forum?id=r1lUOzWCW.

[12] Z. Liu, P. Luo, X. Wang, and X. Tang. Deep learning face attributes in the wild. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 3730–3738, 2015. https://openaccess.thecvf.com/content_iccv_2015/html/Liu_Deep_Learning_Face_ICCV_2015_paper.html.

[13] M. Seitzer. pytorch-fid: FID score for PyTorch, version 0.3.0, 2020. https://github.com/mseitzer/pytorch-fid.

[14] Stability AI. sd-vae-ft-mse model card and weights, 2022. https://huggingface.co/stabilityai/sd-vae-ft-mse.